

RAG-Augmented Multi-Agent LLM Framework for Explainable Software Vulnerability Detection Using CWE Knowledge Bases

Richa Verma (25MCSS02) Advisor: Dr. Akshay Pandey

A small CWE book

This report adds a lookup book of CWE facts. It continues report 2.

What we did

The simple checker should not invent the words for a weakness. Those words should come from a stored page.

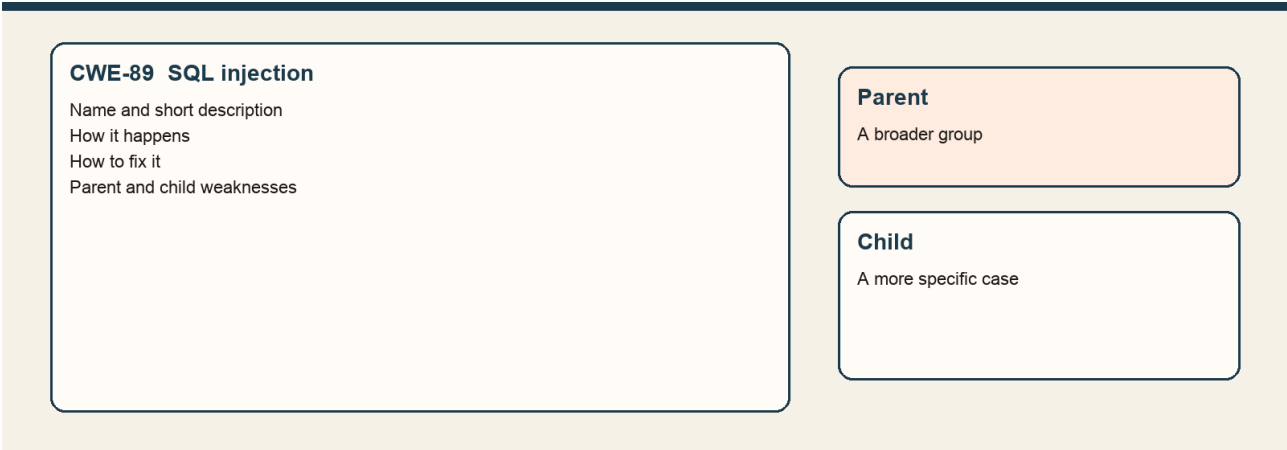
Each page has a name, a short description, related weaknesses, and a fix.

We can look up one id, search by words, and list parents and children.

For example, the CWE-89 page says SQL injection, what goes wrong when user text is stuck into a query, and a safer way to write that query.

The book is a subset we need for this project. It is not the whole CWE catalog. Later answers must use a name from this book. They must not invent one.

Picture



Numbers for this step only

On each page	Why we keep it
Id and name	So the answer uses the official name.
Description	So the note can say what the weakness is.
Parents and children	So a close match can still be explained.
Fixes	So the answer can suggest a repair.

What is next

Next we will search this book from the words in a Java file.

What this report does not claim

This report does not change the detection score from report 2.